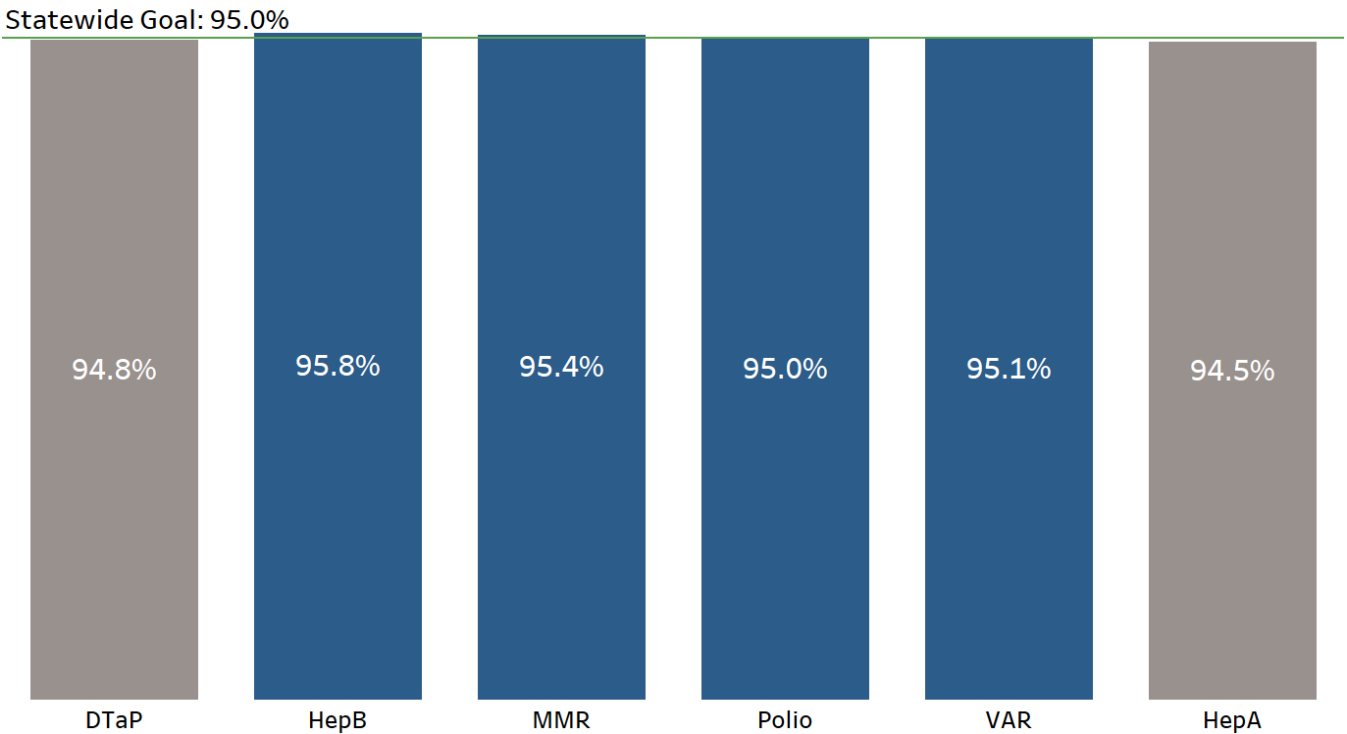


Figure 1. Percentage of **public and private school** kindergarten students who were complete for each vaccine series, compared to the Statewide Goal of 95%. Gray colored bar indicates less than 95% coverage.



Public Schools Statewide Summary

Of the 1,049 public schools that were asked to complete the 2022-2023 Kindergarten Assessment 1,008 (96%) submitted results. Statewide, 93.5% of public-school students were fully immunized at the time of the survey (69,719 of 74,595 students). Students who were not fully immunized fell into the following categories, in decreasing order of prevalence (Note: arrows and corresponding figures represent the percent change greater than 0.1% between 2021-2022 and 2022-2023 assessments):

- Religious Exemption: 2.9% (n=2,154) ↑ 0.7%
- Temporary Certificate: 2.1% (n=1,542) ↑ 0.1%
- Incomplete Record: 1.0% (n=777) ↓ 0.2%
- Missing Record: 0.3% (n=231) ↓ 0.2%
- Medical Exemption: 0.1% (n=107) ↔
- Transfer <30 days: 0.1% (n=65) ↔

Private Schools Statewide Summary

Although our response rate remains relatively stable among private schools, we continue to see a shift towards private schools with an all-time high of 6.4% of all kindergarten enrollment reported in the 2022-2023 Kindergarten Immunization Compliance Assessment. A total of 195 private schools (out of 268 schools; 73%) with kindergarten enrollment completed the 2022-2023 Kindergarten Assessment. Statewide, the survey response rate for private schools increased by 9%. Of those students, only 91.1% (4,644 of 5,097 students) were fully immunized, which represents a 1.7% increase in the fully immunized rate when compared to last year. Unlike the public-school Kindergarten Assessment, the private school survey lacks a central source of enrollment data and thus we are not able to ascertain how many private school students had immunization certificates that

were validated by TennHS. Students who were not fully immunized fell into the following categories, in decreasing order of prevalence (Note: arrows and corresponding figures represent the percent change greater than 0.1% between 2021-2022 and 2022-2023 assessments):

- Religious Exemption: 5.0% (n=253) ↑ 2.8%
- Incomplete Record: 1.6% (n=83) ↓ 1.6%
- Missing Record: 1.1% (n=107) ↓ 1.4%
- Temporary Certificate: 0.9% (n=44) ↑ 0.5%
- Medical Exemption: 0.3% (n=15) ↔
- Transfer <30 days: N/A for private schools based on data collection methods

Trends over Time

In 2022-2023, there was an increase in overall kindergarten assessment enrollment by 0.7% to 79,692. This continues the trend from 2021-2022, where enrollment grew by 7.2% to 79,120 kindergarten students compared with 2020-2021 which saw a 10% decline in overall enrollment during COVID-19. As mentioned, approximately 6.4% of the kindergarten assessment enrollment in this report are from private schools compared to 5.4% in 2021-2022. Since 2015, the percentage of public and private school students who received all vaccines per Tennessee's state requirements for kindergarten entry (i.e., were "fully immunized") has consistently been close to 95%. In recent years (see **Figure 2**) an all-time low of 93.5% of 2022-2023 reported public school kindergarten students were reported as being fully immunized. For a community to remain at low-risk for a highly infectious vaccine-preventable disease outbreak such as measles, this should be above 95%.⁷ With fewer fully immunized kindergarteners statewide, many counties fail to meet the minimum threshold for vaccine-preventable disease immunity. In 2021-2022, the number of counties that met the minimum 95% fully immunized threshold decreased from 72 (approximately 76% of all counties) to 43 (45%) counties in just two years. This trend continued with a sharp decline in 2022-2023 with only 26 counties (27%) meeting this 95% threshold.

During 2022-2023 increased religious exemptions for both public and participating private schools by 0.7% and 2.8% respectively (see **Figure 3**) were described. This continues the upward trend in religious exemptions since 2018, with 2022-2023 being an all-time high with 3% of all kindergarten students have religious exemptions, statewide. In 2022-2023, temporary certificates increased by 0.1% while incomplete records decreased by 0.2%. Medical exemptions increased by 0.1%. These data are concerning for a higher number of children without required immunizations at kindergarten enrollment. An increasing number of kindergarteners who are not fully immunized puts Tennessee schools and communities at risk for vaccine-preventable disease outbreaks and deaths.

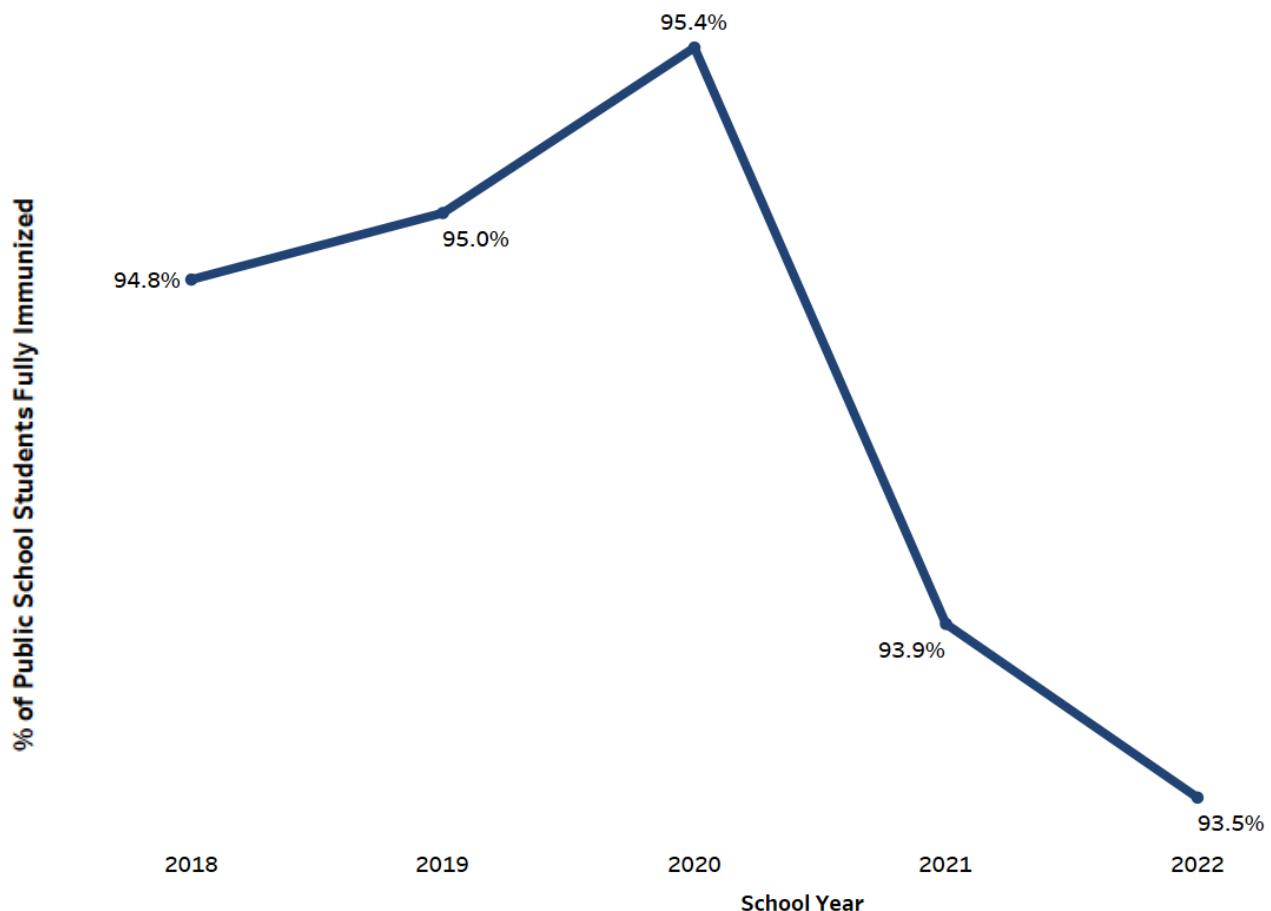
Impact of COVID Pandemic

National estimates have shown a decline in overall vaccination rates following the United States declaration of a national state of emergency due to the SARS-CoV-2 pandemic. This analysis and report describe the impact of the pandemic upon childhood immunization rates in Tennessee. Since pandemic restrictions have been lifted, overall kindergarten survey enrollment increased in 2021-2022 and 2022-2023 school years. After the pandemic started, the number of counties with public school students meeting the 95% fully immunized threshold declined with only 45% (43 of 95) meeting the minimum for vaccine-preventable disease immunization. In 2022-2023, we recorded further decline to 27% (26 of 95) of counties meeting the 95% fully immunized threshold. Of the 34 counties with private school responses to the 2021-2022 survey, only 29% (10) exceed 95% of students

⁸ <https://globalhealthnow.org/2019-12/myth-about-herd-immunity>

fully immunized. For private schools, we saw an increase from 89.4% of students fully immunized to 91.1% which represents a contrast to decreases in vaccination coverage overall. Since the onset of the pandemic, we have observed a decline overall in fully immunized kindergarten students in Tennessee. This decline contrasts with an increase in children who are fully immunized by their 2nd birthday observed in the most recent [Immunization Status Survey Of 24-Month-Old Children in Tennessee](#). Actions to increase compliance with recommended childhood immunizations is needed to decrease the risk of future vaccine-preventable disease outbreaks.

Figure 2. Percentage of **public-school** students fully immunized, 2018-2022



The percentage of all TN kindergarten students who were not fully immunized, by category, over time, are shown in **Figure 3**. The SARS-CoV-2 pandemic resulted in barriers to obtaining immunizations as well as administrative challenges in the schools responsible for enforcing the state's immunization requirements. This resulted in an increase in incomplete immunization records in contrast to previous years (see **Figure 3, gold line**). The trend reversed in 2022-2023 with a decrease of 0.02%. In 2021-22 and 2022-23 school years, temporary certificates (see **Figure 3, blue line**) returned to pre-pandemic levels. The percentage of students claiming a religious exemption increased from 2.3% to 3.0% in 2022-2023 (see **Figure 3, red line**). This year represents the single largest annual increase of 0.7% resulting in a record high number of religious exemptions. The percentage of kindergarten students claiming medical exemptions from a vaccine rose slightly by 0.1% (see **Figure 3, grey line**). The percentage of missing records decreased by 0.2% to 0.4% (not shown in figure). The number of students who lack complete immunization records required for enrollment (incomplete certificate, medical exemption, religious exemption, temporary certificate, missing record) has reached an all-time high of 6.7% allowing for only 93.3% with a complete immunization on record for enrollment. The increases in

High immunization coverage rates are critical to avoiding vaccine-preventable disease (VPD) outbreaks (e.g., measles, pertussis). A VPD outbreak may occur when the community or population is inadequately immunized. To remain at low risk for a VPD outbreak the immunization rate should remain at or above 95% to control highly infectious diseases such as measles [according to the CDC](#). In the maps above and tables below, we categorized an immunization rate of at least 95% as “meets immunization goal” (green), an immunization rate of 90% to 94.9% as “needs improvement” (yellow), and an immunization rate below 90% as “places the community at increased risk” (red).

For detailed Kindergarten Assessment results at the county, school districts, and school levels, please see the *Public-School Districts (SD) by County Summary* section below.

Table 1. Percentage of **public and private school** students fully immunized, total # students, and the number of schools with students fully immunized within each color category, by county, in descending order by “% students fully immunized” (Color Key: <90%; 90-94.9%; 95-100%)

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County Fully Immunized	# Schools with 90-94% Students in County Fully Immunized	# Schools with 95-100% Students in County Fully Immunized
HOUSTON	99.0%	100	0	0	2
CLAY	98.9%	90	0	0	2
DYER	97.5%	472	0	1	5
WAYNE	97.0%	167	0	0	3
COCKE	96.9%	391	0	2	8
BENTON	96.6%	146	0	0	3
JEFFERSON	96.6%	494	0	1	7
DICKSON	96.4%	548	0	3	5
SEQUATCHIE	96.3%	162	0	0	1
MACON	96.3%	376	0	0	3
MARION	96.2%	346	0	2	5
LEWIS	96.2%	158	0	0	1
HENRY	96.2%	316	0	1	3
OBION	96.0%	351	0	1	5
SCOTT	95.8%	330	0	2	4
CUMBERLAND	95.7%	489	0	3	6
STEWART	95.5%	134	0	1	1
CHEATHAM	95.5%	467	1	2	4

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County Fully Immunized	# Schools with 90-94% Students in County Fully Immunized	# Schools with 95-100% Students in County Fully Immunized
MONROE	95.4%	455	0	1	5
CROCKETT	95.3%	214	0	2	3
MEIGS	95.3%	127	0	1	1
ANDERSON	95.1%	823	1	9	7
SULLIVAN	95.1%	1,437	3	8	13
SMITH	95.1%	244	0	2	3
OVERTON	95.0%	242	0	2	3
HARDEMAN	95.0%	260	0	3	3
COFFEE	94.9%	681	1	4	7
MARSHALL	94.8%	403	1	1	1
FRANKLIN	94.8%	382	1	4	3
HENDERSON	94.7%	418	2	2	4
WHITE	94.7%	318	0	2	4
LINCOLN	94.7%	374	2	2	3
HAWKINS	94.6%	467	3	0	7
MCMINN	94.6%	578	1	4	5
CARROLL	94.6%	333	1	0	4
CANNON	94.6%	129	0	1	2
CARTER	94.5%	495	2	4	6
WEAKLEY	94.4%	287	1	2	2
HUMPHREYS	94.4%	233	0	2	2
RHEA	94.3%	370	0	3	2
CHESTER	94.3%	246	0	2	1
TIPTON	94.3%	808	1	2	5
WASHINGTON	94.2%	1,302	4	6	12
MCNAIRY	94.1%	341	0	3	2
SHELBY	94.0%	12,276	28	62	92
PUTNAM	93.9%	891	0	6	5

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County Fully Immunized	# Schools with 90-94% Students in County Fully Immunized	# Schools with 95-100% Students in County Fully Immunized
GREENE	93.9%	705	3	4	7
RUTHERFORD	93.8%	4,565	3	30	13
SEVIER	93.8%	1,179	3	6	5
MONTGOMERY	93.8%	3,077	1	18	11
BEDFORD	93.7%	735	1	4	4
UNICOI	93.7%	159	1	3	0
CAMPBELL	93.7%	443	1	2	6
FAYETTE	93.7%	348	2	1	3
LOUDON	93.6%	605	2	2	5
HICKMAN	93.5%	230	0	2	0
CLAIBORNE	93.5%	337	2	2	5
GIBSON	93.5%	612	4	2	4
ROANE	93.4%	473	1	2	4
ROBERTSON	93.4%	792	4	1	7
LAUDERDALE	93.4%	257	1	0	1
LAWRENCE	93.1%	569	2	4	5
WARREN	93.0%	487	4	2	6
LAKE	93.0%	57	0	1	0
KNOX	92.9%	4,867	19	31	26
HAMBLEN	92.8%	810	1	9	3
GILES	92.8%	292	1	3	0
WILSON	92.8%	1,884	5	9	8
HARDIN	92.8%	249	1	5	0
POLK	92.8%	152	0	2	1
GRAINGER	92.7%	219	1	1	2
BLOUNT	92.7%	1,420	7	7	8
SUMNER	92.6%	2,279	10	15	9
DAVIDSON	92.5%	8,108	32	37	55

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% Students in County Fully Immunized	# Schools with 90-94% Students in County Fully Immunized	# Schools with 95-100% Students in County Fully Immunized
PERRY	92.5%	80	0	2	0
BLEDSON	92.5%	106	0	2	1
WILLIAMSON	92.4%	3,221	14	17	13
DEKALB	92.0%	213	1	0	1
VAN BUREN	91.9%	62	0	1	0
GRUNDY	91.9%	136	1	2	2
DECATUR	91.9%	123	0	2	0
MORGAN	91.9%	209	1	2	2
MADISON	91.8%	1,142	4	9	5
HANCOCK	91.5%	71	0	1	0
JACKSON	91.5%	94	0	2	0
BRADLEY	90.8%	1,154	8	7	4
HAMILTON	90.3%	4,020	24	20	17
JOHNSON	90.1%	202	4	0	1
HAYWOOD	90.0%	190	1	0	1
MAURY	89.4%	1,162	7	7	2
PICKETT	88.9%	54	1	0	0
TROUSDALE	86.7%	113	1	0	0
UNION	84.8%	480	4	2	0
FENTRESS	82.2%	219	4	0	0
MOORE	80.0%	60	1	0	0

Public and Private School Summaries by County and School District

The following tables and graphs describe data: 1) at the county-level, featuring information for both public and private schools within the county; and 2) at the school district level, featuring information for all schools within the school district. Data are presented by county in alphabetical order, with school districts nested under each respective county in alphabetical order, and schools nested under each respective school district in alphabetical order.

Below are descriptions of each vaccine type shown in the county tables and graphs.

- Diphtheria-Tetanus-Pertussis vaccine (DTaP)⁸
- Hepatitis A Virus vaccine (HAV)⁹
- Poliomyelitis vaccine (IPV or OPV)
- Measles, Mumps, Rubella vaccine (MMR)
- Hepatitis B Virus vaccine (HBV)
- Varicella (chickenpox) vaccine (or credible history of disease)

Below are descriptions of each category shown in the county tables and graphs. All table percentages on a given row are the percentage of the “Total Students” number on that row.

County	Tennessee county name. School districts are grouped into their corresponding county.
School District	Department of Education school district name. Schools are grouped into their corresponding school district. Private schools are listed under a “private” district name within each county where applicable.
School Name	School name, listed alphabetically within each school district.
Total Students	Total number of enrolled kindergarten students at the time of immunization record assessment.
Fully Immunized	Number and percentage who meet state school entry requirements by being fully immunized.
Religious Exemption	Number and percentage with a valid religious exemption.
Medical Exemption	Number and percentage with a valid medical exemption to one or more vaccines.
Temporary Certificate	Number and percentage with a valid temporary certificate.
Excluded Transfer	Students exempt from assessment because records are unavailable due to a recent transfer from another TN school (within 30 days of the assessment). N/A for private schools based on data collection methods.

⁸ Diphtheria-Tetanus toxoid vaccine (DT), if medically indicated

⁹ <https://www.tn.gov/health/cedep/immunization-program/ip/immunization-requirements/childcare-12th-grade-immunization-requirements.html>